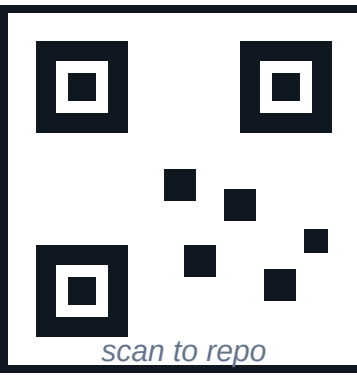


Version-Aware Normalization of Findings Against MITRE ATT&CK v19



Typed Data Models and Revocation-Resolving Technique Lookup for Security Tooling

Pooja Kiran

Security Engineering • AI / Agent Security • Cloud Security

MIT • Python 3.12 • HEAD 756bf03 • verified 2026-09-26

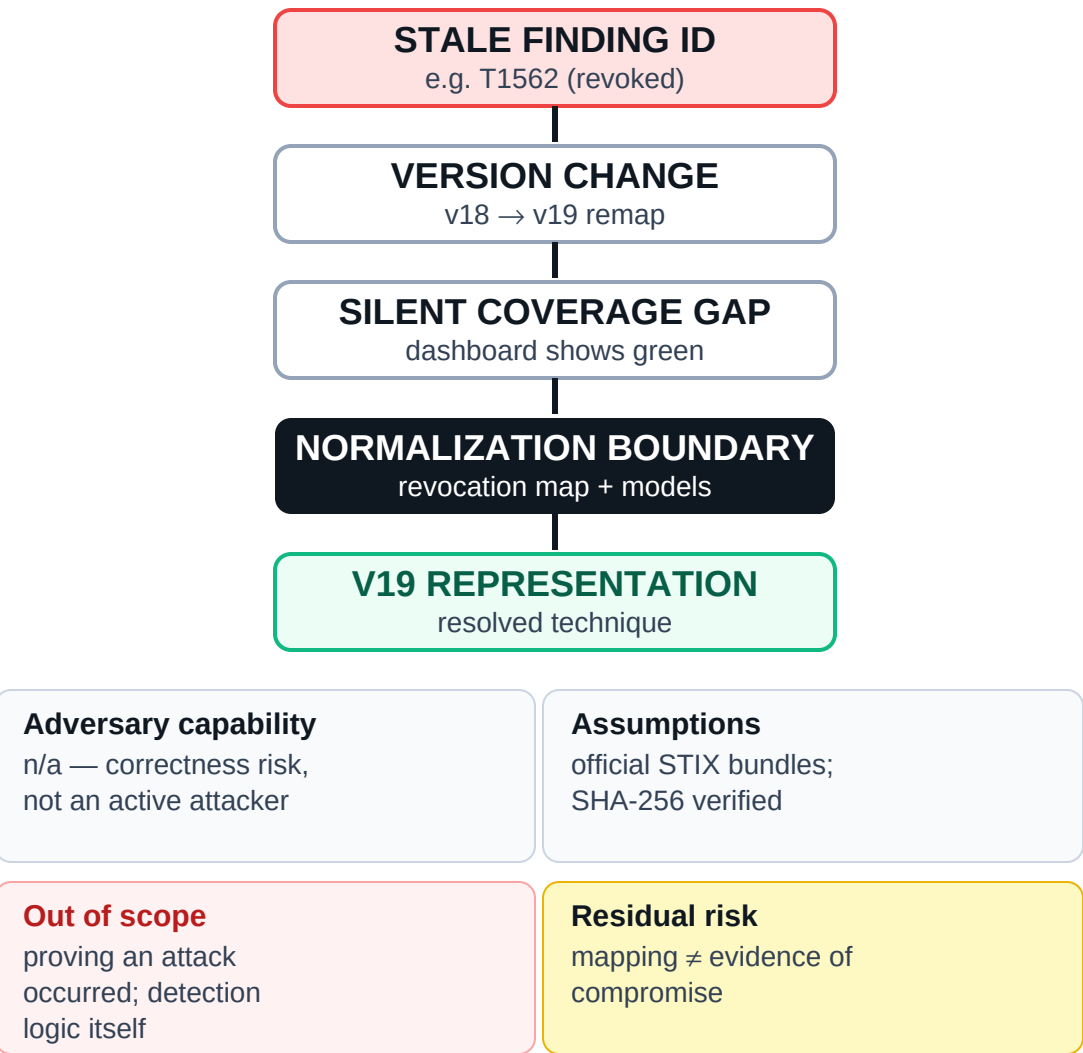
RESEARCH QUESTION

Can security tooling keep mapping findings to ATT&CK across version churn when technique IDs are revoked, renamed, or remapped?

01 • THE FAILURE MODE

ATT&CK v19 revoked 22 technique IDs, renamed a tactic, added TA0112, and introduced 48 techniques. A SIEM rule referencing T1562 now points at a dead ID (replaced by T1685) — the coverage dashboard shows green while a whole tactic is unmonitored. Detection tooling needs to absorb this churn safely.

02 • THREAT MODEL



03 • OBJECTIVE

Determine whether typed models + a revocation map can resolve deprecated ATT&CK IDs to v19 and give O(1) lookup for tooling.

- 01 – Pydantic models per object type
- 02 – Revocation map v18→v19
- 03 – O(1) indexes (id/tactic/platform)
- 04 – SHA-256-verified STIX download

HERO • VERSION-AWARE NORMALIZATION

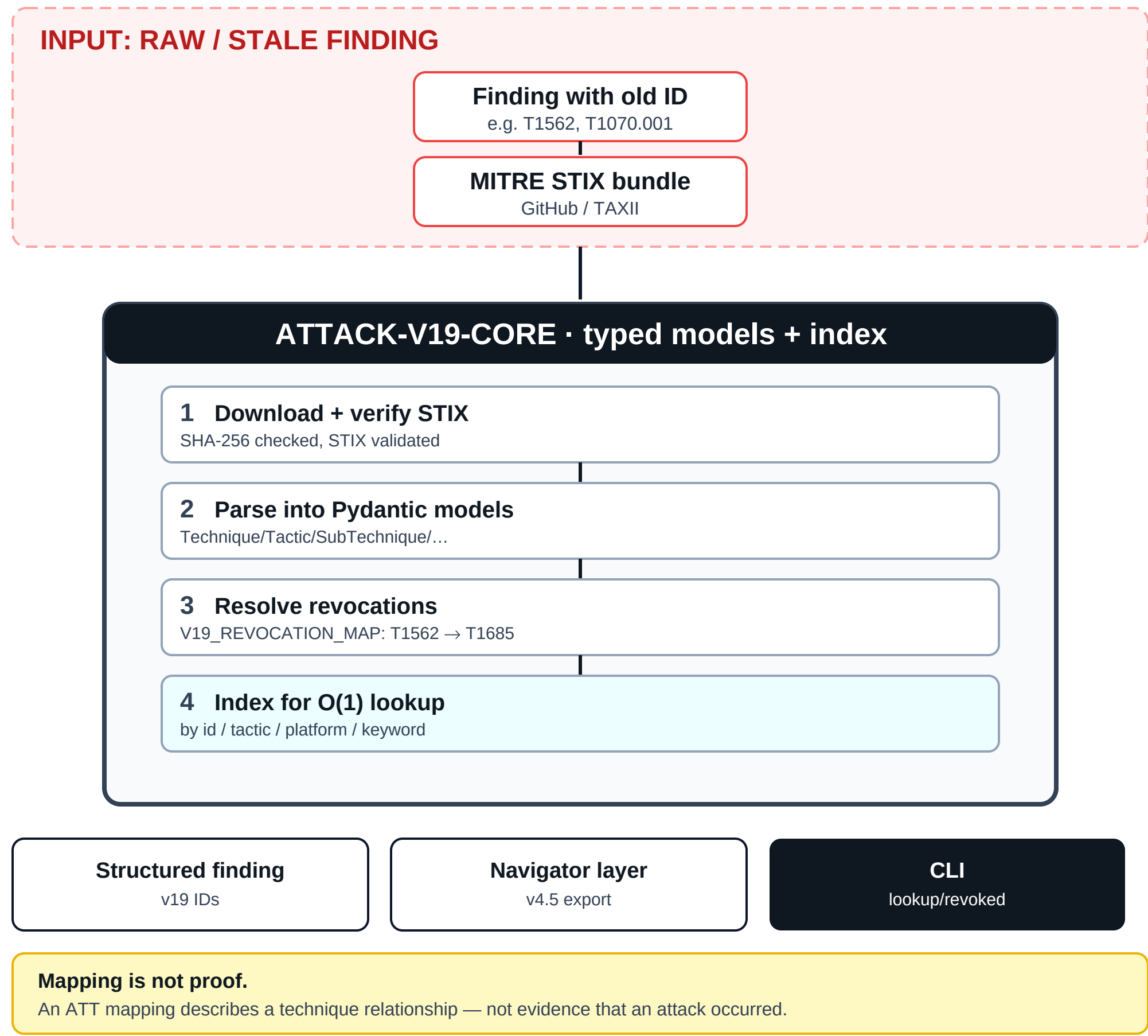
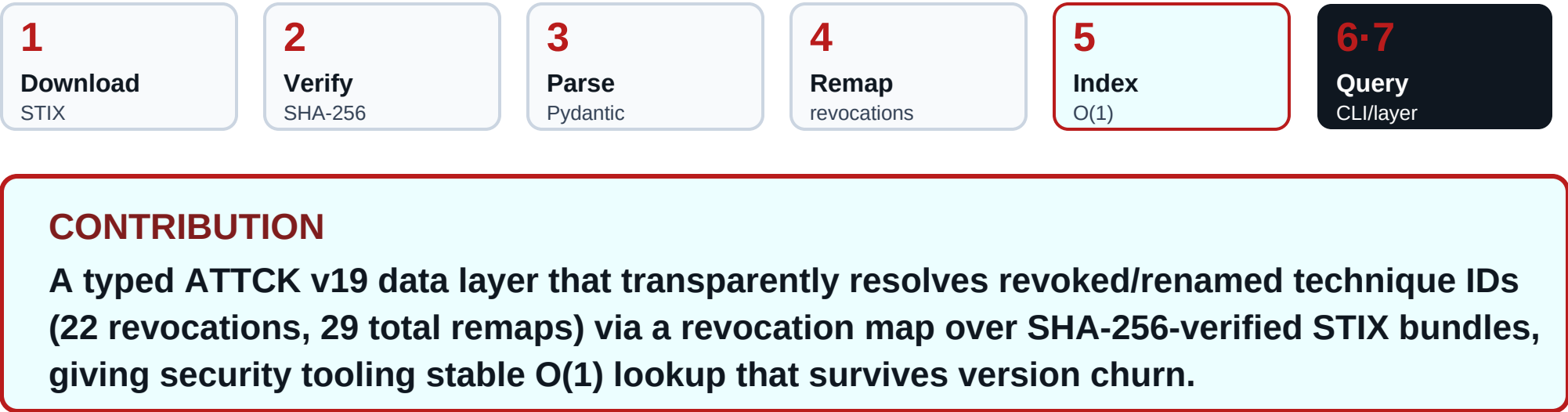
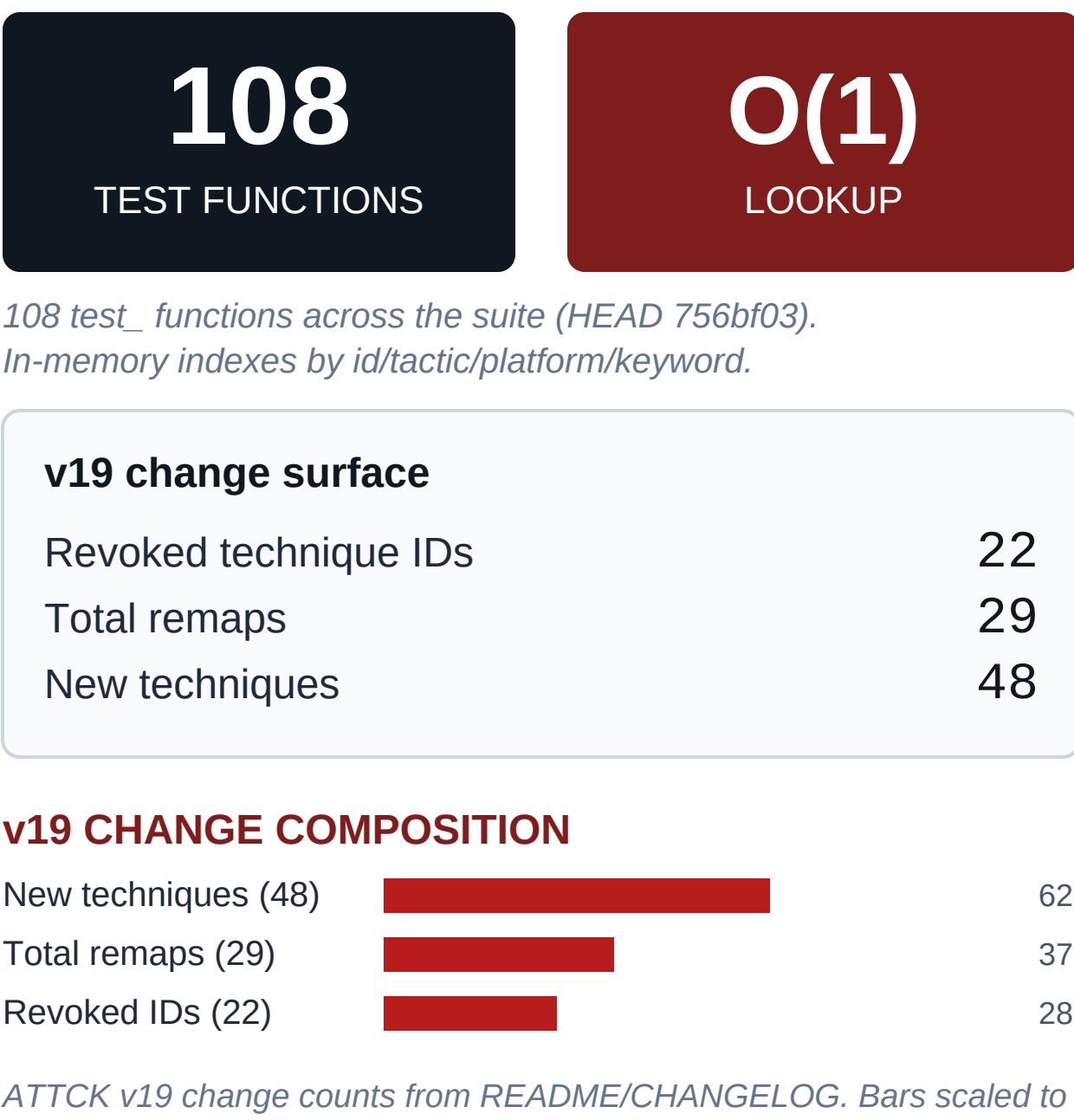


Figure 1. Stale finding IDs are resolved to ATTCK v19 via a revocation map over SHA-256-verified STIX, then indexed for O(1) lookup.

METHOD • RESOLUTION LIFECYCLE



RESULTS • CURRENT LOCAL VALIDATION



INTEGRITY MECHANISM
STIX bundles are SHA-256 verified on download. attack_core is a deprecated shim (DeprecationWarning); attack_v19_core is canonical.

NOT ESTABLISHED BY THIS REPO
That any mapped finding reflects a real attack. Detection logic correctness (this is a data layer).

SECURITY BOUNDARY

PROTECTS / DETECTS

- stable ID resolution
- revocation remap
- typed validation
- SHA-256 STIX
- O(1) lookup

DOES NOT ESTABLISH

- attack occurrence proof
- detection logic
- real-time telemetry
- threat scoring

MAPPING ≠ OCCURRENCE

LIMITATIONS

- 1. A data/normalization layer, not a detector.
- 2. ATTCK mapping does not prove an attack occurred.
- 3. Depends on official STIX bundle availability.
- 4. Deprecated attack_core shim still present (v20 removal).
- 5. Coverage of future versions needs map updates.

FUTURE WORK

- Automated v19→v20 revocation-map generation.
- Remove deprecated attack_core shim.
- TAXII live-sync mode.
- Confidence-scored mapping heuristics.
- Broader ecosystem adapter integration.

CONCLUSION

The library gives security tooling typed ATTCK v19 models and transparent revocation resolution over verified STIX, with O(1) lookup that survives version churn. It normalizes references — it does not detect attacks or prove that any occurred.

REPRODUCE

```
pytest tests/
python -m attack_core lookup T1685
```